

CIT649: STATISTICAL ANALYSIS AND VISUALIZATION-2025 FALL

Group D Project

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0. DATA READING

Read in the dataset and analyze (skim through) the data variables.

Code:

```
load("AAD.RData")
str(AAD)
summary(AAD)

# Optional (recommended):
# library(skimr)
# skim(AAD)
```

What to check:

- Variable types (numeric vs categorical).
- Missingness (NA counts) and impossible values.
- Whether required fields exist (e.g., Outcome, Antibiotic.class, D1/D6 Shannon, D1/D6 Chao1, Jaccard distance).

I. DESCRIPTIVE STATISTICS:

1- Summarizing the data and calculation of the mean, median, minimum, maximum, first and third quartile

Code:

To calculate the mean, median, minimum, maximum, first and third quartile for all the variables:

```
summary(AAD)
```

or write the code for each needed item separately. For example:

```
mean(AAD$D1.Shannon.diversity ,na.rm=TRUE)
median(AAD$D1.Shannon.diversity ,na.rm=TRUE)
min(AAD$D1.Shannon.diversity ,na.rm=TRUE)
max(AAD$D1.Shannon.diversity ,na.rm=TRUE)
quantile(AAD$D1.Shannon.diversity ,na.rm=TRUE,c(0.25,0.75))
```

Definitions:

Statistic	Definition
Mean	Sum of values / total number of values
Median	If n is odd: $(n + 1)/2$; if n is even: average of the two middle values
1st quartile (Q_1)	$0.25(n)$: 25% of the data $< Q_1$, 75% of the data $> Q_1$
3rd quartile (Q_3)	$0.75(n)$: 75% of the data $< Q_3$, 25% of the data $> Q_3$

Summary results:

	Mean	Median	Minimum	Maximum	Q_1	Q_3
D1 Shannon diversity	3.25	3.39	0.13	4.47	2.90	3.77
D6 Shannon diversity	2.86	3.07	0.07	4.46	2.53	3.48
D1 Chao1 diversity	200.84	189.96	25.14	552.93	138.46	247.91
D6 Chao1 diversity	174.42	169.00	36.15	422.75	118.18	222.03
D1.D6.Jaccard.distance	0.65	0.66	0.24	0.95	0.54	0.79

Interpretation:

1. Shannon Diversity:

- Baseline microbial diversity is moderate to high in most patients.
- The median being slightly higher than the mean suggests left-skewness.
- Regarding day 6, Both mean and median decreased compared to Day 1, indicating an overall reduction in microbial diversity after antibiotic exposure.
- The lower minimum (0.07) suggests that some patients experienced loss of diversity by Day 6.
- The downward shift of the IQR confirms that the decline is systematic, not driven by outliers.

2. Chao1 Diversity:

- There is substantial inter-individual variability in baseline Chao1 diversity.
- The large maximum indicates some individuals harbor very rich microbial communities.
- Mean > median suggests right-skewness.
- As for day 6, Both central tendency and spread decreased from Day 1, indicating a loss of species richness following antibiotic treatment.

3. **Jaccard Distance:** The narrow difference between mean and median suggests symmetrical distribution, implying consistent microbiome disruption across individuals.

2- For the categorical variable existing, calculate a frequency table

```
table(AAD$Antibiotic.class)
```

Antibiotic.class	FQN	OBL	PBL
Count	56	111	168

```
table(AAD$Outcome)
```

Outcome	AAD	CDI	ND
Count	22	5	308

Interpretation:

Antibiotic class:

- The study population includes patients exposed to three antibiotic classes:
 - PBL: 168 patients
 - OBL: 111 patients
 - FQN: 56 patients
- PBL is the most frequently prescribed antibiotic class, accounting for the largest proportion of patients.
- FQN is the least represented group, comprising approximately one-third of the size of the PBL group.

Outcome:

- The clinical outcomes are distributed across three categories:
 - ND (No Diarrhea): 308 patients.
 - CDI (*Clostridioides difficile* infection): 5 patients.
 - AAD (Antibiotic-Associated Diarrhea): 22 patients.
- ND is the most common outcome, indicating that the majority of patients (92%) did not develop diarrhea.
- CDI is the least frequent outcome (1.5%), while AAD occurred in about 7% of cases.

3- Calculate the correlation coefficient (D1 Shannon and D6 Shannon) and (D1 Chao D6 Chao)

The correlation coefficient measures the strength of the linear relationship between 2 variables. It ranges between -1 to 1, the negative sign means when one variable increases the other decreases while the positive sign means when one variable increases the other increases and vice versa. The closer the result to 1 the stronger the correlation, 0 means there is no relationship between the 2 variables. Different methods can be used like Pearson and Spearman.

Pearson (default method):

```
cor_1 <- cor(AAD$D1.Shannon.diversity, AAD$D6.Shannon.diversity,
             use="complete.obs")
cor_1
```

0.22

This is a weak positive relationship between D1 Shannon and D6 Shannon.

```
cor_2 <- cor(AAD$D1.Chao1.diversity, AAD$D6.Chao1.diversity, use=
             "complete.obs")
cor_2
```

0.30

This is a weak positive relationship between D1 Chao and D6 Chao.

Spearman: Computes Pearson correlation on the ranks of the data rather than raw values. No requirement for normality, relationship must be monotonic, not necessarily linear, and it is much less sensitive to outliers than Pearson.

```
cor(AAD$D1.Shannon.diversity, AAD$D6.Shannon.diversity, use="
    complete.obs", method="spearman")
```

0.23

This is a weak positive relationship between D1 Shannon and D6 Shannon.

```
cor(AAD$D1.Chao1.diversity, AAD$D6.Chao1.diversity, use="complete
    .obs", method="spearman")
```

0.33

This is a weak positive relationship between D1 Chao and D6 Chao.

Interpretation:

The close agreement between Pearson and Spearman coefficients across both diversity metrics indicates that the associations are stable across methods and are not driven by violations of normality or extreme values. However, because all correlation coefficients are below 0.5, the relationships are considered weak and of limited practical significance.

To view the results:

```
library(corrplot)
cormat <- cor(AAD[, c("D1.Shannon.diversity", "D6.Shannon.
    diversity")], use="complete.obs")
corrplot(cormat, method="color", tl.col="black", tl.srt=45)
```

The plot shows the correlation between D1 Shannon diversity and D6 Shannon diversity. The off-diagonal cell (D1 vs D6) is light blue, consistent with the numerical

correlation of $r \approx 0.22$. The color intensity reflects that the correlation is far from strong (far from 1), visually reinforcing the conclusion that the association is weak.

(Insert Figure: Corrplot of D1 vs D6 Shannon here.)

```
cormat <- cor(AAD[, c("D1.Chao1.diversity", "D6.Chao1.diversity")
], use="complete.obs")
corrplot(cormat, method="color", tl.col="black", tl.srt=45)
```

The plot shows the correlation between D1 Chao1 diversity and D6 Chao1 diversity. The off-diagonal cell (D1 vs D6) is light blue, consistent with the numerical correlation of $r \approx 0.3$. The color intensity reflects that the correlation is far from strong (far from 1), visually reinforcing the conclusion that the association is weak.

(Insert Figure: Corrplot of D1 vs D6 Chao1 here.)

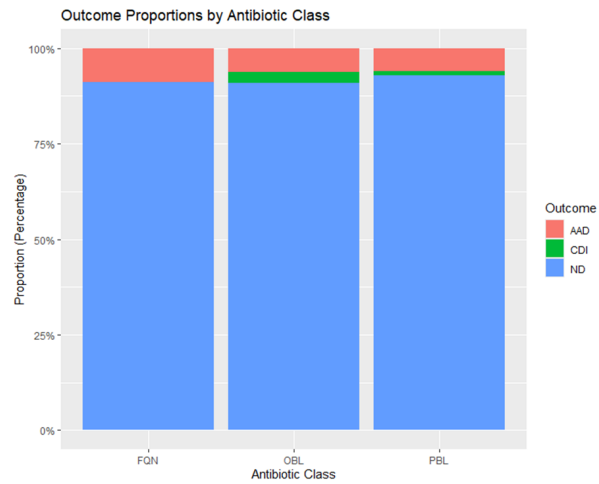


Figure 1: Bar chart of Outcome counts (AAD, CDI, ND)

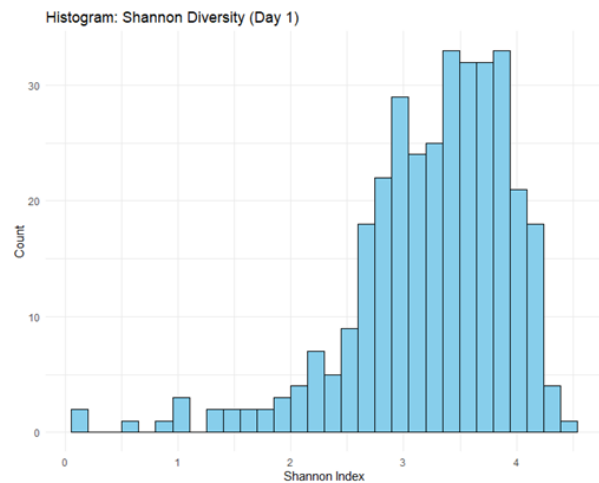


Figure 2: Bar chart / stacked bar of outcome proportions by antibiotic class here.

II. GRAPHICS

Required graphics (Project D):

- Generate a bar chart of a categorical variable for the Outcome (AAD, CDI, ND).
- Generate a bar chart graph with mean Outcome in PBL, FQN, OBL.
- Make a histogram of a continuous variable: D1 Shannon as well as D6 Shannon.
- Make a scatterplot of 2 continuous variables D1 Shannon and D6 Shannon, and add the regression lines for each antibiotic class.
- Make a boxplot of Jaccard distance and separate boxplots per Antibiotics (as factors).

Outcome distribution

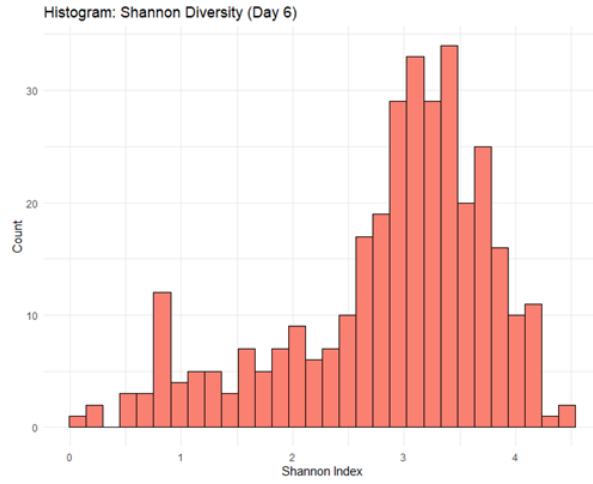


Figure 3: Shannon Diversity (D1 vs D6)

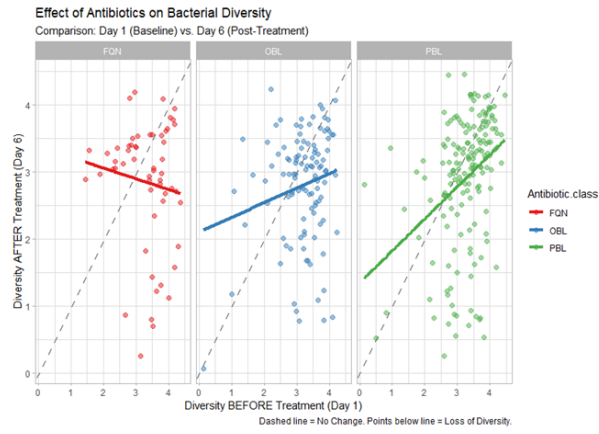


Figure 4: Correlation between baseline and post-treatment diversity. The dashed line represents no change.

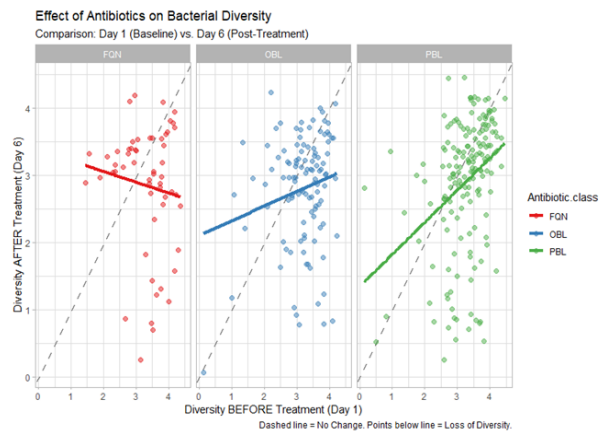


Figure 5: boxplot of D1–D6 Jaccard distance by Antibiotic.class

III. OUTLIER DETECTION

Explore the data for any existing outliers, identify them (do NOT remove them if found).
What do you think?

Result: No outliers were detected using the IQR method.

Interpretation:

- The Jaccard distance is bounded between 0 and 1.
- The IQR calculation produced an upper fence exceeding 1 (e.g., around 1.16), which is outside the possible range.
- Since the statistical upper bound exceeds the maximum possible value (1.0), it is mathematically impossible to have upper outliers under this rule.
- The dataset exhibits high variability, which is expected in biological responses to antibiotics; the observed variation likely reflects real heterogeneity rather than measurement error.

IV. TESTING FOR NORMALITY / HOMOSCEDASTICITY

Requirement: Testing for normality / homoscedasticity (for all features mentioned above).

Normality: check using two methods.

Homoscedasticity: check using two methods.

What do you think?

Normality (two methods)

We checked normality for the main continuous features used above (D1 Shannon, D6 Shannon, D1 Chao1, D6 Chao1, and D1–D6 Jaccard distance) using:

- **Method 1 (Formal):** Shapiro–Wilk test. Example output indicates strong evidence of non-normality for D6 Shannon (very small p-value).
- **Method 2 (Visual):** Q–Q plots and histograms. Q–Q plots show noticeable tail deviations from the straight line, which is consistent with the Shapiro–Wilk result.

Homoscedasticity (two methods)

We assessed constant variance (homoscedasticity) relevant to group comparisons and regression:

- **Method 1 (Formal):** Bartlett test (group variance equality). For example, Bartlett test for Jaccard by antibiotic class did not show strong evidence of unequal variances (p-value > 0.05).
- **Method 2 (Visual):** Residuals vs Fitted plot (regression diagnostic). This plot checks whether residual spread changes across fitted values.

What do we think?

- Several microbiome features (especially Shannon at D6) deviate from normality.
- With a large sample size, parametric procedures are often fairly robust, but assumption results should still be reported.
- For CDI vs ND comparisons, CDI sample size is extremely small (n=5), so both normality assessment within CDI and statistical power are limited; results should be interpreted cautiously.

V. STATISTICAL INFERENCE

Calculate the 90%, 95%, 99% confidence interval for the means of Jaccard distance for each antibiotic class.

Vectors were created containing the Jaccard distance for each Antibiotic.

```
PBL <- AAD[AAD$Antibiotic.class=="PBL",]$D1.D6.Jaccard.distance
OBL <- AAD[AAD$Antibiotic.class=="OBL",]$D1.D6.Jaccard.distance
FQN <- AAD[AAD$Antibiotic.class=="FQN",]$D1.D6.Jaccard.distance

# Checking for missing data
sum(is.na(PBL))
sum(is.na(OBL))
sum(is.na(FQN))
```

There was no missing data in the three vectors.

The CIs were calculated using this function:

```
summary_stats <- function(x, conf.levels=c(0.90, 0.95, 0.99)) {
  n <- length(x)
  mean_x <- mean(x)
  se_x <- sd(x) / sqrt(n)

  CI_df <- do.call(rbind, lapply(conf.levels, function(cl) {
    alpha <- 1 - cl
    t_crit <- qt(1 - alpha/2, df=n-1)
    ci_lower <- mean_x - t_crit * se_x
    ci_upper <- mean_x + t_crit * se_x
    data.frame(conf.level=paste0(cl*100, "%"),
               lower=ci_lower, upper=ci_upper)
  }))
  return(list(mean=mean_x, SE=se_x, n=n, CIs=CI_df))
}

summary_stats(PBL)
summary_stats(OBL)
summary_stats(FQN)
```

Results

	PBL	OBL	FQN
Mean	0.64	0.67	0.65
n	168	111	56
SE	0.0129	0.013	0.02
90% CI (Lower)	0.621	0.650	0.620
90% CI (Upper)	0.663	0.694	0.688
95% CI (Lower)	0.617	0.646	0.613
95% CI (Upper)	0.667	0.699	0.695
99% CI (Lower)	0.608	0.637	0.600
99% CI (Upper)	0.676	0.707	0.708

Interpretation

We are 90% confident that the true mean Jaccard distance for PBL antibiotics lies between 0.621–0.663, for OBL antibiotics lies between 0.650–0.694, and for FQN antibiotics lies between 0.620–0.688.

We are 95% confident that the true mean Jaccard distance for PBL antibiotics lies between 0.617–0.667, for OBL antibiotics lies between 0.646–0.699, and for FQN antibiotics lies between 0.613–0.695.

We are 99% confident that the true mean Jaccard distance for PBL antibiotics lies between 0.608–0.676, for OBL antibiotics lies between 0.637–0.707, and for FQN antibiotics lies between 0.600–0.708.

Conclusion

These inferences indicate that the estimated mean Jaccard distances are stable and precise, with uncertainty appropriately quantified by the confidence intervals. For each group, the confidence intervals at all confidence levels are relatively narrow and show substantial overlap, suggesting that any differences in mean Jaccard distance between groups are unlikely to be large.

When the level of confidence increases, the interval width increases, reflecting increased uncertainty bounds around the same central estimate.

VI. HYPOTHESIS TESTING:

Hypothesis 1: Chao/Shannon at Day 6 is different between CDI vs ND

Question: We hypothesize that Chao/Shannon at day 6 is different between CDI vs ND. Assuming normality and homoscedasticity, test this hypothesis using the statistical hypothesis framework. Assess whether assumptions are met.

(A) Shannon at Day 6 (reported output available)

$$H_0 : \mu_{\text{CDI}} = \mu_{\text{ND}} \quad \text{vs} \quad H_A : \mu_{\text{CDI}} \neq \mu_{\text{ND}}$$

Test used: Two-sample t-test (CDI vs ND) on D6 Shannon diversity.

Reported result (from output):

- $t \approx 0.364$, $df \approx 311$, $p \approx 0.716$
- 95% CI for mean difference: approximately $[-0.644, 0.936]$
- $\text{Mean}(\text{CDI}) \approx 3.050$ and $\text{Mean}(\text{ND}) \approx 2.904$

Decision and conclusion: At $\alpha = 0.05$, we fail to reject H_0 . The data do not provide statistically significant evidence that D6 Shannon differs between CDI and ND in this dataset.

Assumptions check (normality + homoscedasticity):

- Normality: Shapiro–Wilk indicates strong deviation from normality for D6 Shannon; Q–Q plot shows tail deviations.
- Homoscedasticity: for CDI vs ND, variance equality is difficult to verify reliably because CDI has very small sample size ($n = 5$).

(B) Chao1 at Day 6 (same framework; run and insert output)

```
t.test(D6.Chao1.diversity ~ Outcome,
      data = subset(AAD, Outcome %in% c("CDI","ND")))
```

(Insert the test output and interpretation here using the same structure as Shannon.)

Hypothesis 2: Jaccard distance is different between OBL vs FQN (assuming heteroscedasticity)

$$H_0 : \mu_{\text{OBL}} = \mu_{\text{FQN}} \quad \text{vs} \quad H_A : \mu_{\text{OBL}} \neq \mu_{\text{FQN}}$$

Test used: Welch two-sample t-test.

Reported result (from output):

- $t \approx -0.752$, $df \approx 102.69$, $p \approx 0.454$
- 95% CI: approximately $[-0.0666, 0.0300]$
- $\text{Mean}(\text{FQN}) \approx 0.654$, $\text{Mean}(\text{OBL}) \approx 0.672$

Conclusion: At $\alpha = 0.05$, we fail to reject H_0 . There is no statistically significant evidence of a difference in mean Jaccard distance between OBL and FQN.

Assumptions: Welch t-test handles heteroscedasticity. Jaccard is bounded (0,1), so normality may not hold perfectly; interpreting the CI and effect size is important.

Hypothesis 3: Jaccard distance differs across all antibiotic classes (ANOVA + post-hoc)

$$H_0 : \mu_{\text{PBL}} = \mu_{\text{OBL}} = \mu_{\text{FQN}} \quad \text{vs} \quad H_A : \text{At least one mean differs}$$

Reported ANOVA result (from output):

- $F \approx 1.253$, $p \approx 0.287$

Post-hoc (Tukey) result: all adjusted p -values are > 0.25 .

Conclusion: At $\alpha = 0.05$, we fail to reject H_0 . There is no statistically significant evidence that mean Jaccard distance differs across antibiotic classes in this dataset.

VII. LINEAR REGRESSION

Question: Fit a linear regression to the data and interpret the regression coefficient (for one hypothesis above). Calculate and interpret a 95% confidence interval of the regression slope (bonus). Estimate the average Jaccard distance change with changing antibiotics (bonus).

Model

We fit:

$$\text{Jaccard}_{D1 \rightarrow D6} = \beta_0 + \beta_1 I(\text{OBL}) + \beta_2 I(\text{PBL}) + \varepsilon,$$

with FQN as the reference group.

Interpretation of coefficients (from output):

- Intercept ($\beta_0 \approx 0.6539$): estimated mean Jaccard distance for FQN.
- β_1 (OBL vs FQN, ≈ 0.0183): OBL mean is about 0.018 higher than FQN (not statistically significant).
- β_2 (PBL vs FQN, ≈ -0.0119): PBL mean is about 0.012 lower than FQN (not statistically significant).
- Model fit: $R^2 \approx 0.0075$ indicates antibiotic class explains less than 1% of variation in Jaccard.

95% confidence intervals (bonus)

From reported coefficient CIs:

- OBL vs FQN: approximately $[-0.032, 0.069]$
- PBL vs FQN: approximately $[-0.059, 0.035]$

Both intervals include 0, consistent with non-significant coefficients.

Estimated average Jaccard by antibiotic (bonus)

Using the fitted coefficients:

- FQN: $\hat{\mu} \approx 0.6539$
- OBL: $\hat{\mu} \approx 0.6539 + 0.0183 \approx 0.6722$
- PBL: $\hat{\mu} \approx 0.6539 - 0.0119 \approx 0.6420$

Diagnostics linked to your visuals

- Normal Q-Q plot: shows tail deviations, suggesting residual non-normality (common with bounded outcomes).
- Residuals vs Fitted: helps assess whether variance changes across fitted values; fitted values cluster by group because the predictor is categorical.

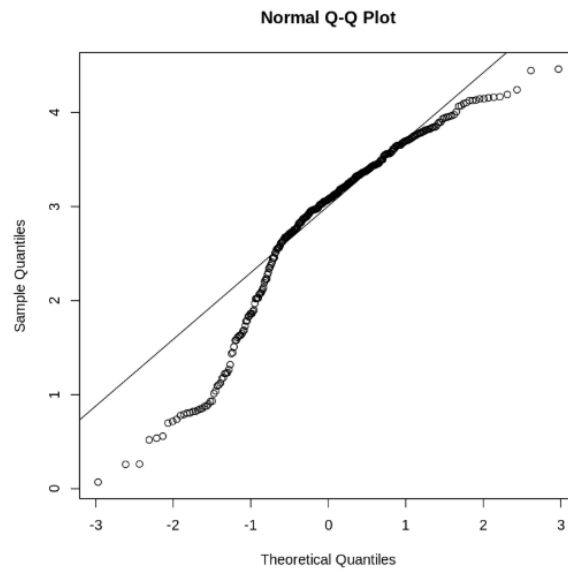


Figure 6: Normal Q-Q plot

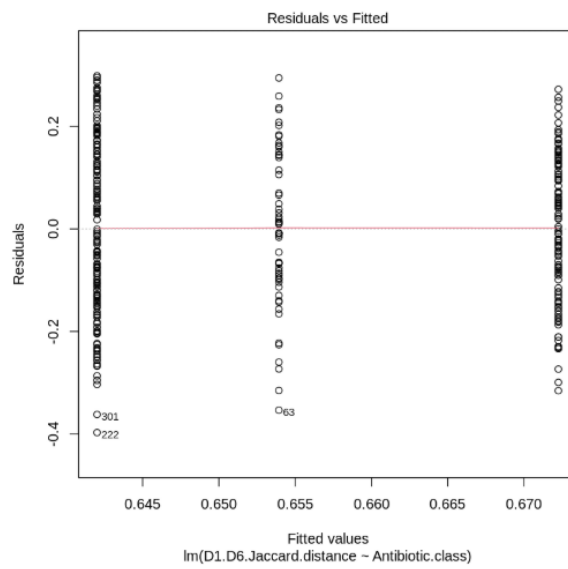


Figure 7: Residuals vs Fitted